A. Sizing code verification: % errors

1. Design point output aircraft verification % errors (using book takeoff gross weight)

For the table below, the takeoff gross weight stays fixed at the true value for each aircraft. The L2000 used the SR-71 as a baseline since they both use a titanium skin. Airfoil data was only available for the B737-200, B707-120, and C5A. The airfoil on the B737-200 was applied to the subsonic airliners without airfoil data. heavier airliners typically have a higher lift to drag ratio which is why the B777-200er and A320-200 are excluded from this list (no airfoil data for those aircrafts). Raymer empty weight fudge factors were applied to each common aircraft class and aircraft outer skin material.

- TOGW = Takeoff gross weight
- W_e = Empty weight
- W_f = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

Table 1: Percent errors of predicted performance parameters relative to reference aircraft data (using true takeoff gross weight).

Aircraft	*TOGW*	W_e	W_f	*Alt*	*R*	*L/D*	*TSFC*
Supersonic fighter equations							
XB-70	Book	3.4	8.7	Book	Book	3.2	0.5
SR-71	Book	4.5	4.2	Book	Book	2.6	2.7
L2000	Book	2.9	5.0	Book	Book	7.2	19.6
Supersonic transports							
Concorde	Book	0.9	2.3	Book	Book	0.8	2.0
TU-144D	Book	2.6	6.6	Book	Book	3.1	2.5
Subsonic transports							
B737-200	Book	1.9	8.4	Book	Book	1.5	2.9
2000 EX	Book	2.2	7.6	Book	Book	4.7	0.5
B707-120	Book	0.5	3.0	Book	Book	0.6	3.2
C5A	Book	5.2	5.2	Book	Book	1.8	2.2
Subsonic general aviation							
CJ3	Book	1.2	2.5	Book	Book	0.7	6.7

2. Geometric wing output verification % errors (using book takeoff gross weight)

The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- b = wingspan
- c_r = root chord length
- Λ = leading edge sweep angle

Table 2: Percent errors of predicted wing geometric variables relative to reference aircraft data (using true takeoff gross weight).

Aircraft	*S*	*b*	c_r
Supersonic fighter equations			
XB-70	2.2	4.3	1.9
SR-71	3.4	1.7	4.6
L2000	0.5	0.3	56.1
Supersonic transports			
Concorde	0.7	0.3	12.2
TU-144	0.3	0.1	1.6
Subsonic transports			
B737-200	1.2	0.6	4.5
2000 EX	3.9	2.0	2.0
B707-120	1.2	0.6	7.3
C5A	2.6	1.3	1.9
Subsonic general aviation			
CJ-3	0.5	0.3	0.3

3. Design point output aircraft verification % errors (takeoff gross weight free to iterate)

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect. The L2000 used the SR-71 as a baseline since they both use a titanium skin. Airfoil data was only available for the B737-200, B707-120, and C5A. The airfoil on the B737-200 was applied to the subsonic airliners without airfoil data. heavier airliners typically have a higher lift to drag ratio which is why the B777-200er and A320-200 are excluded from this list (no airfoil data for those aircrafts). Raymer empty weight fudge factors were applied to each common aircraft class and aircraft outer skin material.

- TOGW = Takeoff gross weight
- W_e = Empty weight
- W_f = Fuel weight
- Alt = design altitude
- R = range
- L/D = Lift to drag ratio
- TSFC = Thrust specific fuel consumption

Table 3: Percent errors of predicted performance parameters relative to reference aircraft data (allowing takeoff gross weight to iterate).

Aircraft	*TOGW*	W_e	W_f	*Alt*	*R*	*L/D*	*TSFC*
Supersonic fighter equations							
XB-70	12.0	8.4	15.2	1.6	1.0	0.7	0.8
SR-71	36.4	20.0	48.6	21.9	5.9	1.7	17.2
L2000	16.9	6.1	26.8	19.2	2.7	3.8	2.9
Supersonic transports							
Concorde	12.1	8.0	17.3	15.6	1.8	15.6	15.6
TU-144D	18.5	13.3	23.7	16.4	0.8	9.9	17.9
Subsonic transports							
B737-200	0.6	2.1	4.2	0.9	3.3	1.4	3.2
2000 EX	0.3	2.5	2.6	2.6	0.1	4.8	0.4
B707-120	4.9	2.6	8.0	17.7	2.9	0.2	8.9
C5A	13.1	13.7	18.2	1.5	6.4	3.2	2.8
Subsonic general aviation							
CJ3	3.6	3.4	4.8	16.5	2.3	9.8	3.7

4. **Geometric wing output verification % errors (takeoff gross weight free to iterate)**

For the table below, the takeoff gross weight is free to iterate which significantly multiplies the magnitude of the percent errors from the tables above due to the snowball effect. The reason for the high error for the root chord length of the Concorde in the table below is because the Concorde has an ogive wing while the formulas in the sizing code are for a trapezoidal and triangular delta wing.

- S = wing area
- b = wingspan
- c_r = root chord length
- Λ = leading edge sweep angle

Table 4: Percent errors of predicted wing geometric variables relative to reference aircraft data (allowing takeoff gross weight to iterate).

Aircraft	*S*	*b*	c_r
Supersonic fighter equations			
XB-70	14.4	10.4	3.8
SR-71	34.2	18.9	16.6
L2000	16.5	8.6	42.2
Supersonic transports			
Concorde	12.7	6.6	17.7
TU-144	18.7	9.8	11.1
Subsonic transports			
B737-200	0.0	0.0	14.3
2000 EX	4.1	2.1	2.1
B707-120	3.6	1.8	4.7
C5A	10.9	5.6	5.1
Subsonic general aviation			
CJ-3	4.1	2.1	2.1

Table 5: Percent errors of predicted performance parameters relative to reference aircraft data with sweep solver turned on

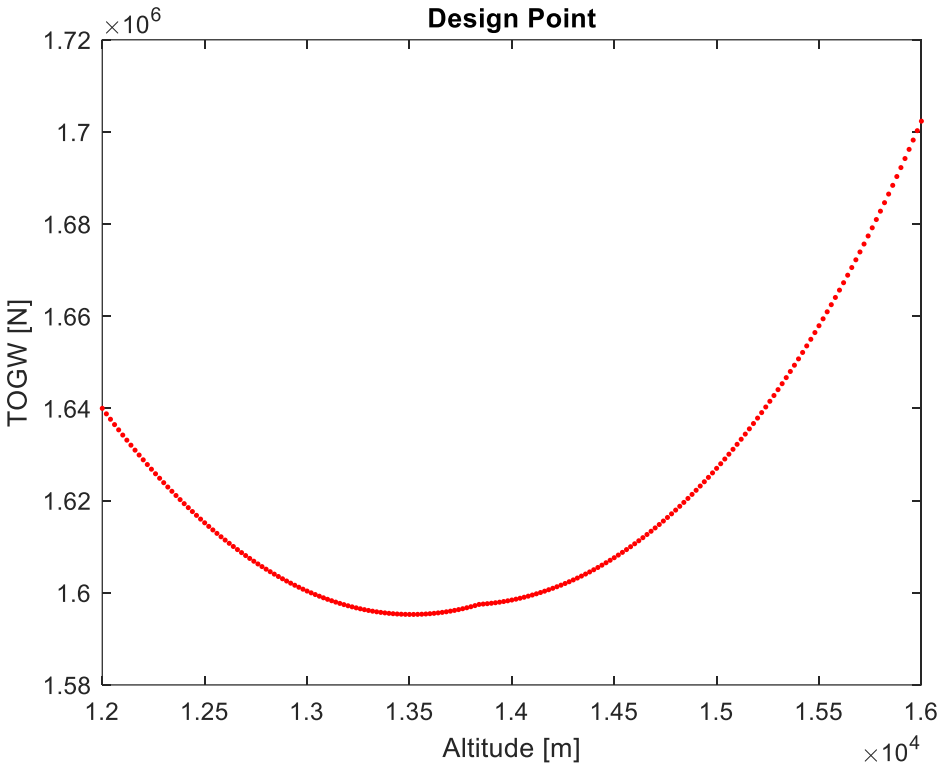
Aircraft	*TOGW*	W_e	W_f	*Alt*	*R*	*L/D*	*TSFC*	*S*	*b*	c_r	Λ
B737-200	Book	6.9	4.1	Book	3.4	3.7	3.5	10.0	35.0	12.5	3.4
B737-200	1.3	6.9	4.1	2.7	3.4	3.7	3.5	10.0	35.0	12.5	3.4
B707-120	Book	0.2	2.3	Book	0.9	0.9	3.1	0.4	0.2	6.9	2.0
B707-120	4.5	0.4	1.6	2.6	2.1	1.8	1.9	35.7	3.5	12.5	2.0
C5A	Book	4.6	6.1	Book	6.3	1.8	2.1	3.7	1.8	2.4	4.1
C5A	9.3	4.4	1.9	3.1	3.7	4.0	4.2	20.5	19.3	0.6	4.1

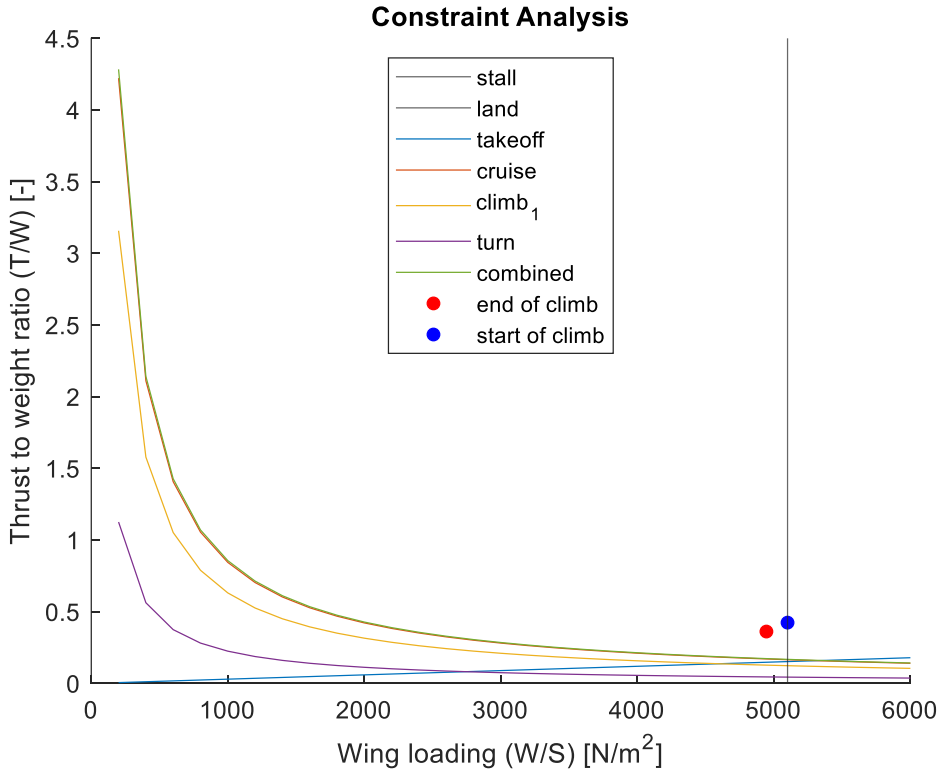
Sweep was only solved for on the B737-200, B707-120, and C5A since those are the only aircrafts with airfoil data. XFOIL was used to obtain the minimum pressure coefficient of each airfoil.

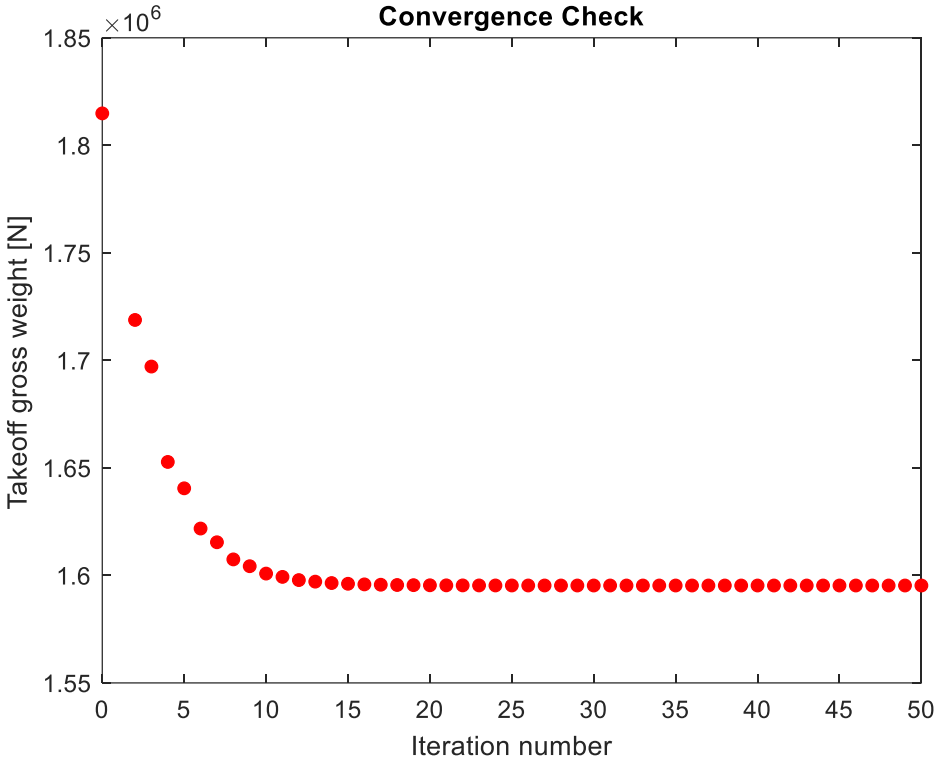
5. Example supersonic aircraft sizing code outputs: Concorde

Model status: These scripts are conceptual/preliminary engineering models. Several aerodynamic, propulsion, structural, and thermal relationships are analytical or empirical approximations. Results should not be interpreted as certification-level aircraft predictions. The purpose of the models is to investigate multidisciplinary aircraft-design relationships.

- TOGW=1595297.935052
- W_e=710513.129024
- Range=6309080.687433
- Lift to drag ratio=6.025545
- Drag_per_engine=50206.664552
- S=312.793704
- Alt=13500.000000
- V=590.136037
- V_a=264.730531
- c_r=22.776837
- c_t=3.416525
- b=23.883433
- Sweep=55.000000
- C_l=0.090052
- C_l_incompressible=0.000000
- alpha=0.062532
- C_d_0 (use in airfoil opt. code) = 0.009243
- Max thickness sweep (sweep_xc)=0.553210
- Re_wing_root=233280508.594617
- CPR=15.500000
- BR=0.000000
- FPR=0.000000
- A_inlet=0.951000
- Thrust_required=50206.664552
- TIT=1400.000000
- TSFC=0.000029
- S_error=12.688429
- b_error=6.559339
- c_r_error=17.654242
- TOGW_error=-12.097532
- We_error=-7.970223
- Wf_error=17.338933
- alt_error=15.625000
- R_error=-1.759366
- L/D_error=15.608617
- TSFC_error=15.551775


Design Point
×10^6
1.72
1.7
1.68
1.66
1.64
1.62
1.6
1.58
TOGW [N]
1.2
1.25
1.3
1.35
1.4
1.45
1.5
1.55
1.6
Altitude [m]
×10^4


Constraint Analysis
stall
land
takeoff
cruise
climb 1
turn
combined
end of climb
start of climb
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
Thrust to weight ratio (T/W) [-]
0
1000
2000
3000
4000
5000
6000
Wing loading (W/S) [N/m^2]


Convergence Check
×10^6
1.85
1.8
1.75
1.7
1.65
1.6
1.55
Takeoff gross weight [N]
0
5
10
15
20
25
30
35
40
45
50
Iteration number

6. Methods used in the sizing code in order:

a) Aerodynamics methodology to obtain the lift to drag ratio

- Obtain subsonic airfoil drag polar data of various airfoils using airfoiltools.com
- Begin the altitude sweep and TOGW convergence for-loops
- Calculate ambient atmospheric properties using [1]
- Obtain the minimum pressure coefficient using XFOIL for the sweep solver (only done for B737-200, B707-120, C5A). The wing sweep solver uses [2-5].
- Solve for the wing area using CL_max using [6].
- Solve for average weight during cruise and use it to size the aircraft.
- Calculate wing aspect ratio for a delta wing, wing area, and span, Oswald efficiency factor [7-8], root chord length, sectional and overall lift curve slopes. The wing area is calculated from the max takeoff gross weight and the sea level stall speed and air density.
- Calculate engine drag using the frontal area of each engine.
- Calculate sectional and 3D lift curve slopes using [9-10].
- Calculate the compression lift using the shock properties using [11].
- Solve for the fuselage length (if desired) using the fuel volume as the fuselage volume minus 10 meters of length for non-fuel volume.
- Solve for the coefficient of skin friction using empirical relations using [12-17]
- Solve for the fuselage and wing parasite drag coefficient [18], zero-lift wave wing drag coefficient [19-20], and fuselage wave drag coefficient [21]. Together, these make up the total zero-lift drag coefficient. The zero-lift wing wave drag formula uses a simplifying relation that depends on total airfoil thickness.
- Solve for the coefficient of lift using the mean weight and compression lift (if activated). For the subsonic case, the airfoil drag polar data is used to obtain the coefficient of lift and drag (adding induced drag in separately later).
- Solve for angle of attacks using the lift-curve slope and coefficient of lift.
- Solve for the induced drag coefficient [22-23].
- Solve for the coefficient of wave drag due to lift using a simplifying expression [24].
- Solve for the total drag. For the supersonic case, the total drag is zero lift + induced + wave due to lift + engine drag. For the subsonic case, since drag polar data is used, The total drag is the drag polar drag + induced drag + fuselage parasite drag.
- Calculate high speed supersonic aircraft surface temperature if applicable [25].
- Calculate the thermal protection system weight (supersonic) that depends on the weight of the insulation type material (if not already included in the Raymer empty weight formulas) and the weight of the active cooling jacket.
- Finally, calculate the lift to drag ratio that is fed into the Breguet fuel mass formula [30] for the convergence of the TOGW at each altitude.

b) Propulsion methodology to obtain the thrust specific fuel consumption (TSFC)

- An if statement is created that controls which set of equations to solve: for the ideal engine or for an existing engine. For the ideal engine, a series of for-loops are created that solves for engine variables like bypass ratio, compressor pressure ratio, and turbine inlet temperature that must produce a thrust equal to the drag of the aircraft per engine.
- The inlet, compressor, fan, and turbine total temperatures and pressures are solved for using [26-18].
- The fuel flowrate per air flowrate fraction is solved for using the turbine inlet temperature [derived].
- The exit temperature is solved for and is used to calculate the engine core air velocity [derived].
- A for loop is created to solve for the sea level mass flow rate. The kinetic energy formula (Ke_dot=.5*m_dot*Ve^2) is used to help solve for m_dot.
- Next, the core thrust is solved for and adjusts the turbine inlet temperature to thrust match the drag of the aircraft. If the total turbine inlet temperature becomes maxed out, the afterburner is applied for the remaining thrust.
- The afterburner exit velocity is solved for by using the specified total afterburner temperature which solves for the fuel fraction of the afterburner required to generate the remaining thrust that the core itself cannot generate without the afterburner [derived].
- Next, the thrust specific fuel consumption is solved for.

c) Structural methodology to obtain weights

- Raymer's approximations are used to determine all the different weight components of the empty weight using [29].
- Next, the empty weight is found by adding the listed weights together plus the skin, thermal protection system (supersonic), and payload weights.

d) Performance and trajectory methodology for climb and descent

- The total fuel required for the descent weight is found first by starting with the landing weight and sink rate.
- Next, the Breguet fuel mass is calculated using this weight at the end of cruise to determine the weight at the beginning of cruise [29].
- Next, the total fuel required for the climb phase is calculated by starting with the initial cruise weight and climb rate.
- Finally, the total takeoff gross weight is found by adding the climb weight to the initial cruise weight. This value is the TOGW used for each altitude.

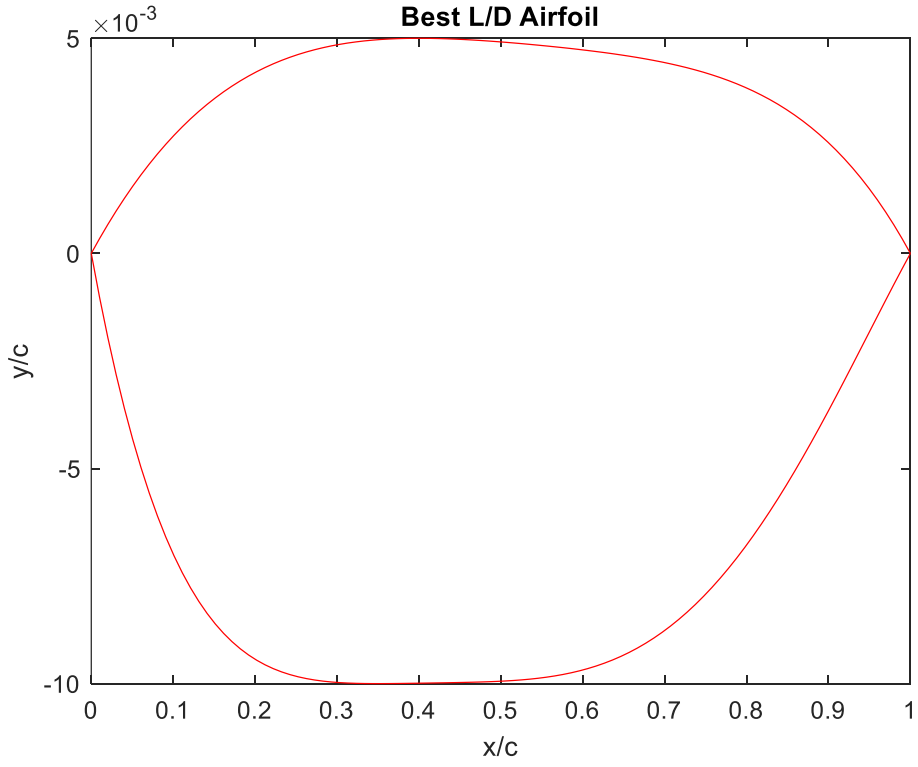
e) Carpet plots (constraint diagrams)

- The wing loading equations for stall and land are specified [derived].

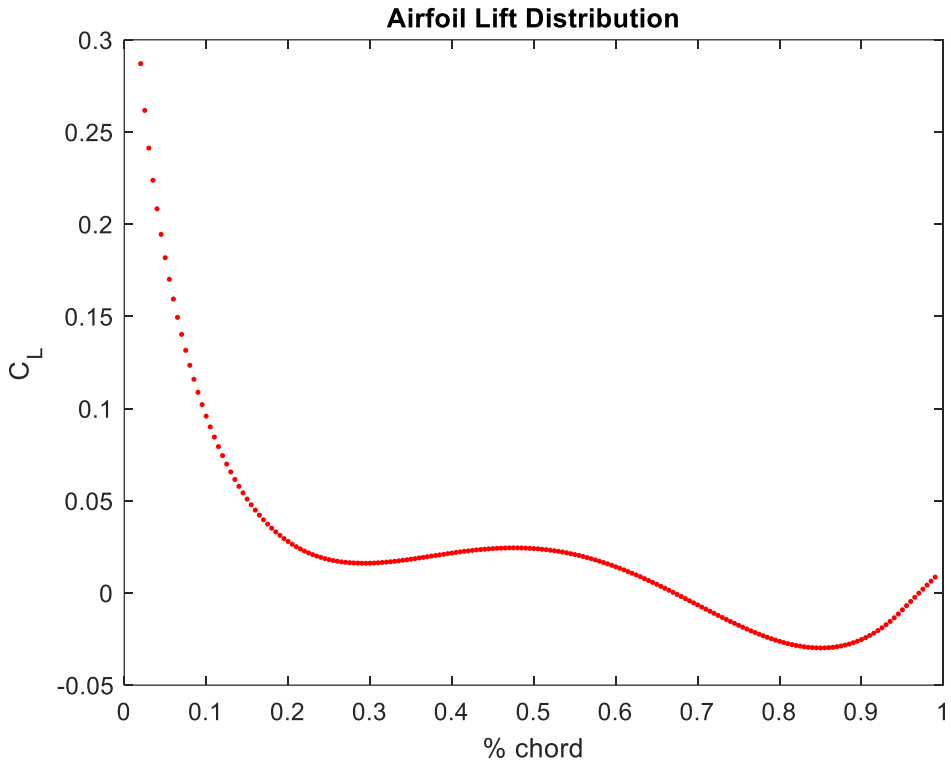
- The thrust to weight ratio for takeoff, climb, cruise, and turn are specified [derived]. These equations use the required thrust. The takeoff distance includes clearing a 50ft obstacle [derived].
- Next, the initial climb and initial cruise design point is plotted against these curves. These two points use the available thrust.
- Next, the upper left space defined by the curves is the area for feasibility.

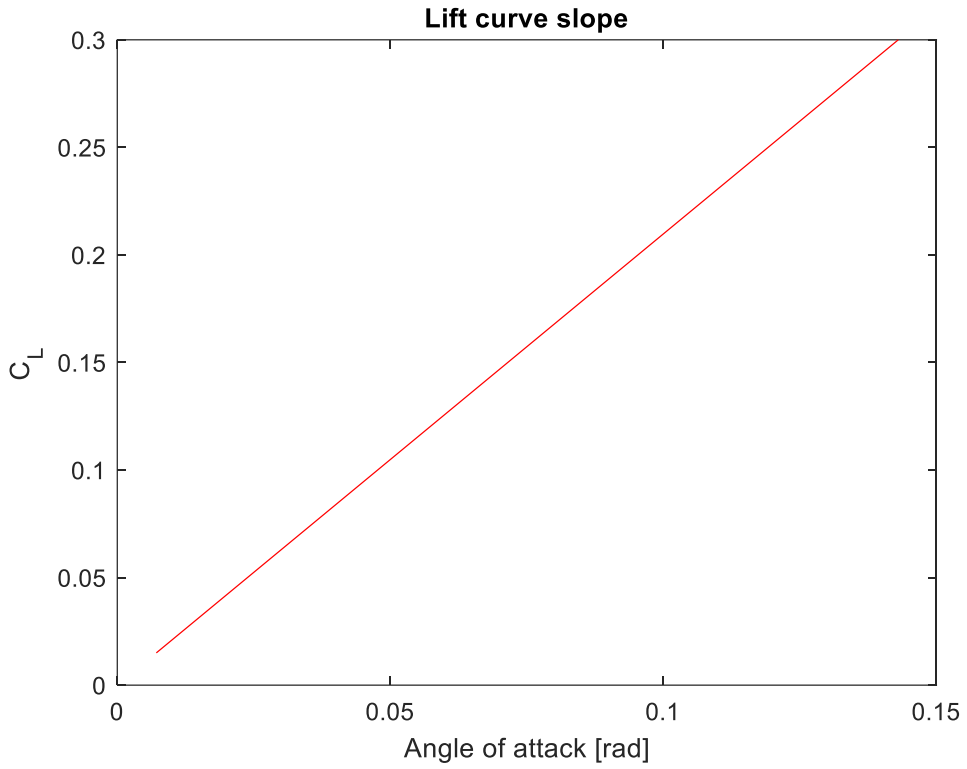
7. Airfoil optimization code outputs for the Concorde flight condition

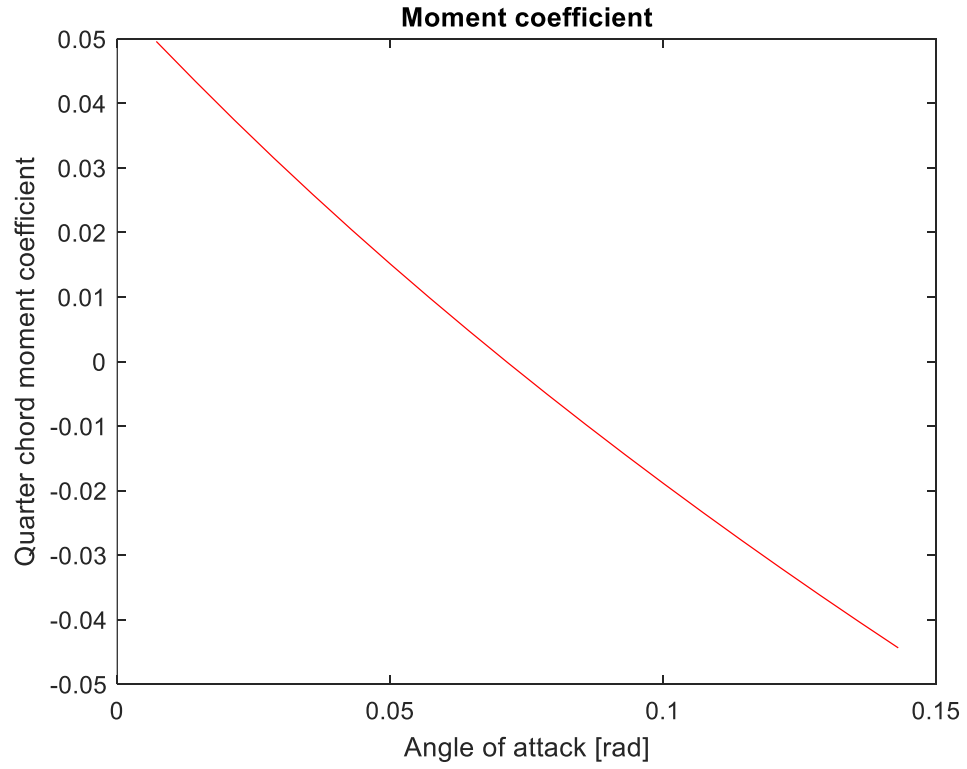
- aoa=0.063182
- x_crest_u=0.400000
- y_crest_u=0.005000
- x_crest_l=0.350000
- y_crest_l=-0.010000
- L/D=8.233306



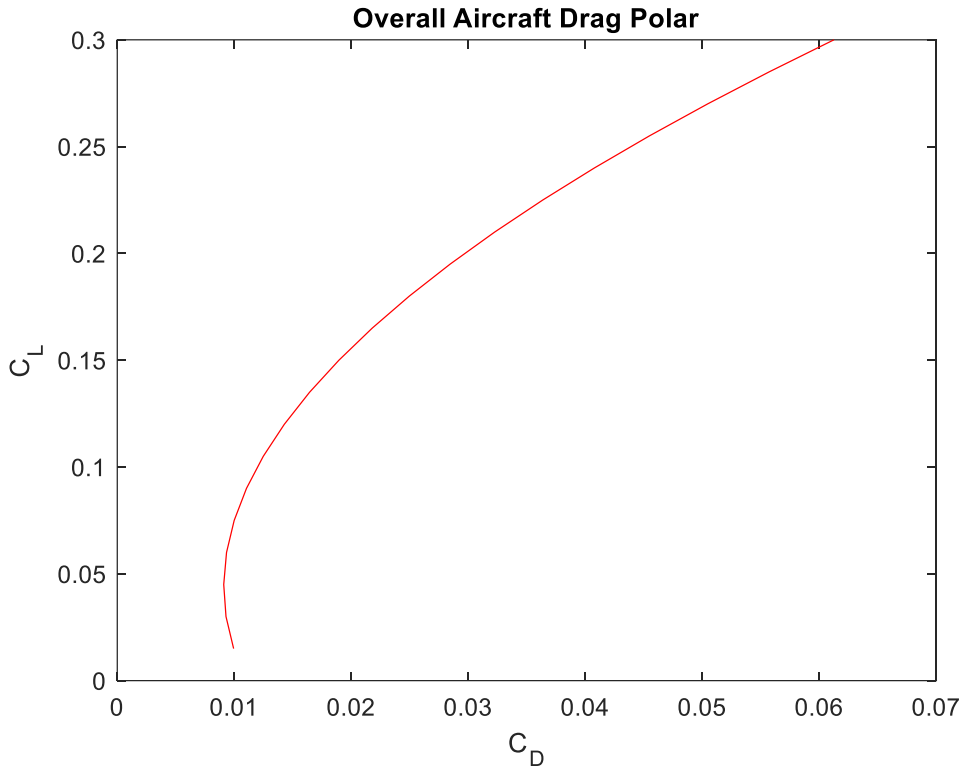
The reason the lift to drag ratio is higher than the actual Concorde lift to drag ratio is because the best airfoil thickness to chord ratio is smaller than the actual Concorde thickness to chord ratio.

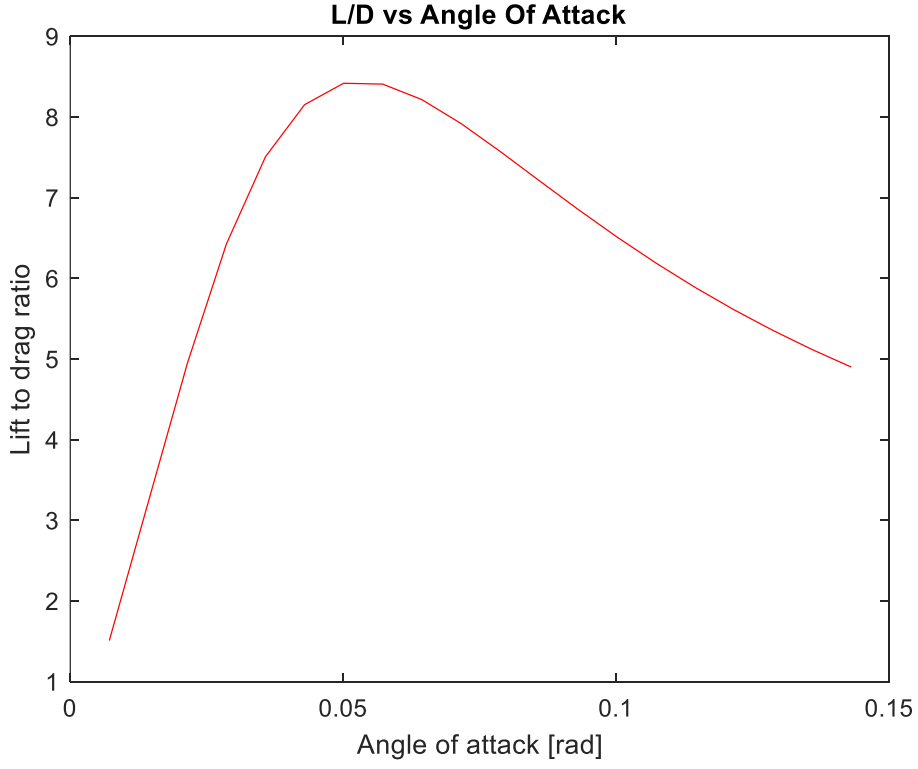

Airfoil Lift Distribution
0.3
0.25
0.2
0.15
0.1
0.05
0
-0.05
C_L
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
% chord


Lift curve slope
0.3
0.25
0.2
0.15
0.1
0.05
0
C_L
0
0.05
0.1
0.15
Angle of attack [rad]


Moment coefficient
0.05
0.04
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
-0.04
-0.05
Quarter chord moment coefficient
0
0.05
0.1
0.15
Angle of attack [rad]

..


Overall Aircraft Drag Polar
0.3
0.25
0.2
0.15
0.1
0.05
0
C_L
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
C_D


L/D vs Angle Of Attack
9
8
7
6
5
4
3
2
1
Lift to drag ratio
0
0.05
0.1
0.15
Angle of attack [rad]

8. Methods used in the airfoil optimization code in order:

- For loops of the max thickness location and max thickness of the upper surface and lower surface air foil are created to iterate for the best combination.
- Next, the coefficients of a polynomial are solved for using the geometric input variables of the airfoil. The subsonic polynomial takes the form y/c = a1*x^(1/2) + a2*x + a3*x^2 + a4*x^3 + a5*x^4. The supersonic polynomial takes the form y/c = a1*x + a2*x^2 + a3*x^3 + a4*x^4 + a5*x^5.
- Next, for the subsonic case, for-loops are created to solve for the A coefficients and zero lift angle of attack of thin airfoil theory [32]. For the supersonic case, shock and expansion relations are used using [35].
- Next, the velocity at each point along the chord is calculated using the A coefficients for the upper and lower surface for the subsonic case [33-34].
- Isentropic relations are then used to calculate the pressure coefficient at each point along the chord using [26].
- Next, the axial and normal force coefficients are solved for and converted into the coefficient of lift and drag with the ratio calculated.
- Next, for the subsonic case, the equations for Stratford's flow separation criteria are solved for using [31].
- Next, the lift to drag ratio is compared for each combination of geometric variables of the airfoil. The highest lift to drag ratio is the output airfoil.

9. References

Aerodynamics Sources

1. **Atmospheric temperature and pressure versus altitude:** NASA Glenn Research Center. "Earth Atmosphere Model." *NASA Glenn Research Center*, https://www.grc.nasa.gov/www/k-12/airplane/atmosmet.html.
2. **Approximation for minimum pressure coefficient:** National Aeronautics and Space Administration. *NASA Technical Report*, p. 10. NASA Technical Reports Server, https://digital.library.unt.edu/ark:/67531/metadc54820/m2/1/high_res_d/19930082349.pdf.
3. **Compressibility correction to minimum pressure coefficient:** *Technical reference on compressibility effects*, p. 545, eq. 11.51. https://dl.icdst.org/pdfs/files4/1080b2835404bf5affd0cd56181ba457.pdf.
4. **Minimum pressure coefficient with wing sweep correction:** *NASA Technical Report*, p. 26, eq. 3-9. NASA Technical Reports Server, https://ntrs.nasa.gov/api/citations/19960003144/downloads/19960003144.pdf.
5. **Equations for critical Mach number:** "C-2: Compressibility Effects." *Computational Simulations for Aerodynamics*, Embry-Riddle Aeronautical University, https://eaglepubs.erau.edu/compsimforaero/chapter/c-2/.
6. **Maximum coefficient of lift (C_L_max):** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.15, p. 268.
7. **Oswald efficiency factor:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eqs. 12.49–12.50, p. 297.
8. **Oswald efficiency factor equation 12.50 and its applicability to sweep angles greater than 30 degrees:** "An Integrated Method for Determination of the Oswald Factor in a Multi-Fidelity Design Environment." *ResearchGate*, https://www.researchgate.net/publication/225025023_An_Integrated_Method_for_Determination_of_the_Oswald_Factor_in_a_Multi-Fidelity_Design_Environment.
9. **3D Lift curve slope (subsonic):** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.6, p. 264.
10. **2D and 3D supersonic lift curve slope, 2D subsonic lift curve slope:** Cook, Michael V. *Flight Dynamics Principles: A Linear Systems Approach to Aircraft Stability and Control*. 2nd ed., eq. 12.14, 12.17, 12.18, p. 330-331 https://repo.poltekbangsby.ac.id/id/eprint/12/1/Flight%20Dynamics%20Principles,%20Second%20Edition%20A%20Linear%20Systems%20Approach%20to%20Aircraft%20Stability%20and%20Control%20by%20Michael%20V.%20Cook%20(z-lib.org).pdf.
11. **Theta-beta-M oblique shockwave relation:** *Wikipedia*, https://en.wikipedia.org/wiki/Oblique_shock.

12. **Wing form factor for parasite drag:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.30, p. 281.
13. **Fuselage form factor for parasite drag:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.31, p. 281.
14. **Nacelle form factor for parasite drag:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.32, p. 281.
15. **Cutoff Reynolds number:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eqs. 12.28–12.29, p. 281.
16. **Coefficient of skin friction:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.27, p. 280.
17. **Coefficient of skin friction that works well for the XB-70:** National Aeronautics and Space Administration. *NASA Technical Report*, eq. 1, p. 7. NASA Technical Reports Server, https://ntrs.nasa.gov/api/citations/19730001588/downloads/19730001588.pdf.
18. **Total parasite drag:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eq. 12.24, p. 279.
19. **Zero-lift wing wave drag:** *Aircraft Design: Synthesis and Analysis*, p. 210.
20. **Zero-lift wing wave drag:** Fitzpatrick, Richard. "Wave Drag." *Fluid Mechanics*, University of Texas at Austin, https://farside.ph.utexas.edu/teaching/336L/Fluid/node212.html.
21. **Sears-Haack minimum body wave drag:** National Aeronautics and Space Administration. *NASA Technical Report*, eq. 42b, p. 23. NASA Technical Reports Server, https://ntrs.nasa.gov/api/citations/19930084002/downloads/19930084002.pdf.
22. **Subsonic induced drag:** NASA Glenn Research Center. *NASA Glenn Research Center*, https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/induced-drag-coefficient/.
23. **Supersonic induced drag:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eqs. 12.56–12.60, pp. 300–302.
24. **Wing wave-drag coefficient for swept wings:** *Supersonic Aerodynamics Reference*, p. 125, first equation. Archived document, Internet Archive, https://web.archive.org/web/20240727010634/https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=8c1039e96f43e3ff287194d39ac5d169f36a7ac0.

Aerothermal Sources

25. **Aerothermal equations and high-speed aircraft heating analysis:** National Aeronautics and Space Administration. *NASA Technical Report*. NASA Technical Reports Server, https://ntrs.nasa.gov/api/citations/19900019499/downloads/19900019499.pdf

Propulsion Sources

26. **Isentropic flow relations:** NASA Glenn Research Center. "Isentropic Flow." *NASA Glenn Research Center*, https://www.grc.nasa.gov/www/k-12/airplane/isentrop.html.

27. **Isentropic efficiency of turbine and compressor:** *ScienceDirect Topics*, sec. 13.5.2, first and second formula. Elsevier, https://www.sciencedirect.com/topics/engineering/isentropic-efficiency.
28. **Turbine efficiency:** *ScienceDirect Topics*. Elsevier, https://www.sciencedirect.com/topics/materials-science/turbine-efficiency.

Structures and Weight Estimation Sources

29. **Empty-weight component estimation:** Raymer, Daniel P. *Aircraft Design: A Conceptual Approach*. 2nd ed., eqs. 15.1–15.58, pp. 397–403.
30. **Fuel mass fraction using the Breguet range equation:** Massachusetts Institute of Technology. "The Breguet Range Equation." *MIT Unified Engineering*, https://web.mit.edu/16.unified/www/FALL/thermodynamics/notes/node98.html.

Airfoil Optimization Sources

31. **Stratford's flow separation criterion:** *CFD-Online*, https://www.cfd-online.com/Wiki/Stratford%27s_separation_criterion.
32. **Thin airfoil theory:** O'Neill, Charles R. "Thin Airfoil Theory." https://charles-oneill.com/projects/iaf.pdf.
33. **Formula for vortex-sheet strength (gamma) and circulation:** Gementzopoulos, Antonios. *Gust Encounter Flow Physics with Applications to Flow Sensing and Control*. 2024, p. 25, first formula. PhD dissertation.
34. **Compressibility correction to circulation and consequently vortex strength (gamma):** Santos, Pedro Daniel da Rocha. *Aero-Structural Wing Analysis Using a Vortex Lattice Method Coupled with an Equivalent Plate Model*. p. 14, eq. 2.19.
35. **Shockwave equations, including total-pressure ratio across a shockwave:** NASA Glenn Research Center. "Normal Shock Wave Equations." *NASA Glenn Research Center*, https://www.grc.nasa.gov/WWW/K-12/airplane/normal.html